Insurance Charges Regression Assignment

1. Identify the problem

The objective is to predict insurance charges using age, sex, body mass index (BMI), number of children, and smoking status.

Problem Identification
Stage 1- Machine Learning
Stage 2 -What type of learning – Supervised Learning
Stage 3 – Regression/Classification – based on output parameters – Regression

2. Basic information about the dataset

The insurance_pre.csv dataset contains 1,338 rows and six columns. Each row represents an individual. Five columns are input features, and charges is the target variable.

3. Pre-processing

I converted the categorical features sex and smoker into numerical indicator columns using one-hot encoding with pandas.get_dummies(). These categories are nominal and do not have an inherent numerical order.

I applied standardization to place numerical features on a comparable scale. Standardization centers values around the mean and scales them by their standard deviation; it does not reduce the underlying differences between individuals.

4. Model development and comparison

I evaluated Multiple Linear Regression, Support Vector Regression (SVR), Decision Tree Regression, and Random Forest Regression. I tested alternative settings for SVR, Decision Tree, and Random Forest and compared their reported test R^2 scores.

Rank	Model	R2_Value
1	***Random Forest***	***0.8919***
2	Support Vector Regression	0.8567
3	Multiple Regression	0.7895
4	Decision Tree	0.7529

5. Detailed Experiment Results

Multiple Regression

Multiple Regression R2 value is **0.7895**

Support Vector Regression

I tested linear, radial basis function (RBF), polynomial, and sigmoid kernels with different values of C. The polynomial kernel with C = 1000 achieved the highest reported test R^2 of 0.85664877

S. No.	Hyper parameter	Linear	Nonlinear (RBF)	Polynomial	Sigmoid
0	Without C	-0.0101027	-0.083382386	-0.0756997	-0.0754292
1	C = 0.01	-0.0888313	-0.089645537	-0.0895683	-0.089565
2	C = 0.10	-0.08096	-0.089074515	-0.0883024	-0.0882699
3	C = 1	0.01010267	-0.083382386	-0.0756997	-0.0754292
4	C = 10	0.46246841	-0.032273294	0.03871622	0.03930714
5	C = 100	0.62887929	0.320031783	0.61795696	0.52761035
6	C = 1000	0.76493117	0.810206485	**0.85664877**	0.28747069

Decision Tree

I tested four criteria, two splitters, and three max_features settings, giving 24 experiments. Experiments 21 and 23 tied for the highest reported test R^2 of 0.7529632. Both used criterion="poisson" and splitter="best", with max_features="sqrt" and "log2", respectively.

SN	Criterion	Splitter	Max Features	R^2 Value
1	squared_error	best	None	0.702487
2	squared_error	random	None	0.7210197
3	squared_error	best	sqrt	0.6830549
4	squared_error	random	sqrt	0.6969298
5	squared_error	best	log2	0.6830549
6	squared_error	random	log2	0.6969298
7	friedman_mse	best	None	0.718117
8	friedman_mse	random	None	0.7140828
9	friedman_mse	best	sqrt	0.6802509
10	friedman_mse	random	sqrt	0.6969298
11	friedman_mse	best	log2	0.6802509
12	friedman_mse	random	log2	0.6969298
13	absolute_error	best	None	0.6707086
14	absolute_error	random	None	0.750635
15	absolute_error	best	sqrt	0.7073084
16	absolute_error	random	sqrt	0.647562
17	absolute_error	best	log2	0.7073084
18	absolute_error	random	log2	0.647562
19	poisson	best	None	0.7372896
20	poisson	random	None	0.7019495
21	poisson	best	sqrt	0.7529632
22	poisson	random	sqrt	0.626101
23	poisson	best	log2	0.7529632
24	poisson	random	log2	0.626101

Random_Forest

I completed 24 Random Forest experiments. Experiment 16 achieved the highest reported test R^2 of 0.89186328, rounded to 0.8919. It used absolute_error, 100 trees, max_features=0.7, max_depth=None, and min_samples_leaf=4.

S.No.	Criterion	n_estimators	max_features	max_depth	min_samples_leaf	R^2 (test)
1	squared_error	50	1	None	1	0.8413
2	squared_error	100	1	None	1	0.8509
3	squared_error	100	0.7	None	1	0.868
4	squared_error	100	1	None	2	0.8584
5	squared_error	100	0.7	None	2	0.8798
6	squared_error	100	1	None	4	0.8329
7	squared_error	100	0.7	None	4	0.8871
8	squared_error	100	1	3	1	0.6051
9	squared_error	100	0.7	3	1	0.8671
10	squared_error	100	1	3	2	0.6076
11	absolute_error	100	1	None	1	0.8533
12	absolute_error	100	0.7	None	1	0.8691
13	absolute_error	100	1	None	2	0.8598
14	absolute_error	100	0.7	None	2	0.8864
15	absolute_error	100	1	None	4	0.8242
16	**absolute_error**	**100**	**0.7**	**None**	**4**	**0.8919**
17	absolute_error	100	1	3	1	0.5464
18	absolute_error	100	0.7	3	1	0.8544
19	absolute_error	100	1	3	2	0.5491
20	absolute_error	100	0.7	3	2	0.8554
21	poisson	100	1	3	1	0.5993
22	poisson	100	1	3	2	0.6042
23	poisson	100	1	5	1	0.7851
24	poisson	100	1	5	2	0.8006

6. Final Model and Justification

I selected Random Forest Regression because it achieved the highest reported test R^2 among the models and settings tested. Its R^2 of 0.8919 exceeded SVR at 0.8567, Multiple Linear Regression at 0.7895, and Decision Tree Regression at 0.7530.

Within the Random Forest experiments, the best absolute_error result of 0.8919 was slightly higher than the best squared_error result of 0.8871. The selected model therefore used the following settings.

Repository link: https://github.com/mmrousan/Assignment-Regression-Algorithm.